

HUONG SEN COMFOR YARN SPINNING COMPANY LIMITED

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Optifil + RS Spindle Monitoring System

for

18 sets Rieter G32 X 1824 ring frame

Embrace Industry 4.0

PINTER GROUP HAS INTRODUCED SEVERAL INNOVATIVE PRODUCTS TO THE MARKET

Throughout its almost one-hundred-year history, the Pinter Group has introduced several innovative products to the market that allowed it to always be on the crest of the wave and become one of the most appreciated leaders in the spinning sector.

Attachments to add value to the yarn, such as equipment to process elastane on ring frame, or equipment to create slub and injected yarn, represent only a part of the rich range of products offered by Pinter Group.

Aware that the Textile Industry is undergoing an important transformation phase, the Italian-Spanish company has decided to support those customers who need digital solutions to embrace Industry 4.0, introducing the EFFIMILL monitoring system in the market.



HOW TO OPTIMIZE PRODUCTIVITY AND LOWER COSTS WITH EFFIMILL

The EFFIMILL monitoring system is made up of a hardware part and a software platform called SmartDragon.

The hardware consist of sensors, detectors, hygrometers and thermometers that collect data directly from the machines.

The SmartDragon software processes all this data and generates useful information to improve the organization in the Spinning, Weaving and Knitting departments.

In order to increase the productivity of a plant, it is necessary to know in detail both the performance of machines and raw materials, as well as the efficiency of the work team.

Only a system like EFFIMILL can provide this essential data in real time, allowing to optimize processes and be competitive in the market.



Sensor System

Optifil – Optical Yarn Detector

Optifil is an individual optical detector to keep monitored the status of each yarn position in real time. It has been developed to be installed on any kind of ring frame for short staple fiber. placed on the ring rail in front of the ring, checking traveller's speed of each individual spindle at real time, so advising in case of yarn breakage and informing when there is an abnormal traveller



speed, to prevent low twist problems. - 1 signal light in every spindle with 5 different colours. - 1 bicolor LED (Red and blue) in every 24 spindles to highlight an abnormal status

Advantages:

- No parts in movement: less maintenance.
- High quality aluminum bars; rust never appears.
- It works with any count and color of yarn
- The only way to ensure 100% spindles without low twist.
- It can be combined with the Roving Stop, to cut the feeding of the roving in case of yarn breakages.
- Short distance dim light with 5 colors and long distance high light with 2 colors
 1. Short distance signal and long distance signal light are separate:
 - a) With long distance high light, easy for operator to discover the breakage area for long frame.
 - b) Short distance light is dim, reduce the power consumption.
 - c) With dim light of short distance signal, the life time of the light can be enlarged significantly.
 - d) Under the situation of more breakages, separate the long/short distance light (so that operator will only see the limited number of high light from long distance) will reduce the stress of the operator and also reduce the light pollution.
 2. Function of short distance light:
 - a) **Red**: breakage; **Blue**: Low twist; **Blinking Blue**: Idle; **Violet**: Rogue; **Yellow**: obstacle/dust; **Green**: Roving stop function
 - b) According to the different color of the same light, workers can tell the condition of the spindle and take action accordingly.
 3. Function of long distance light:

Red: request the help from operator team; **Blue**: request the help from maintenance team.

ST-BOX4 and CT-BOX4 Terminals – Spinning



Terminals were designed for machines with batch control and with the option to enter machine stoppage reasons manually.

The ST-BOX4 terminal is used for monitoring the spinning machines equipped with Optifil sensors, while the CT-BOX4 terminal is used to monitor the preparation machines to provide a complete view of the factory.

The operator interface is a 7" Color Display and to simplify use by employees, it offers graphics and touch-screen capabilities.

These allow management of:

- Current batches/article
- Upcoming batches (ten in preparation and one in spinning)
- Batch changes
- Batch statement (production, efficiency, breakdown trends)
- Time working and time stopped
- Production speed
- Meters and kg produced during shift
- Machine stoppage reasons reported automatically (e.g., doffing, wound on a cylinder, sliver missing)
- Machine stoppage reasons entered manually
- Operator shift
- Breakages Management with alerts for breakages and twists (only for spinning machines)

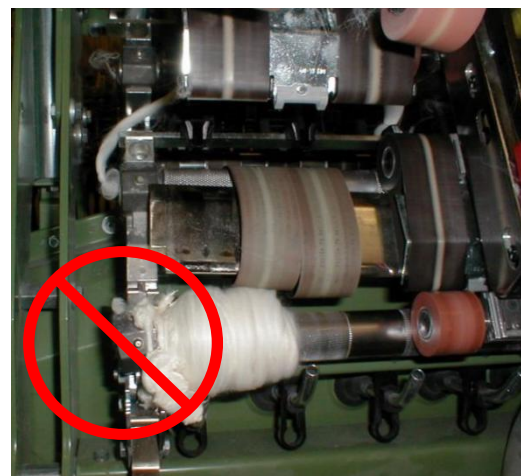
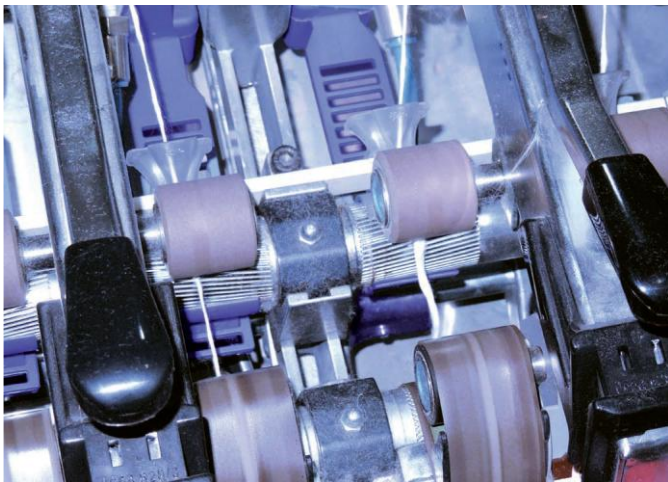
Signaling methods for active breakages

Indicators (signal lights or LCD) are installed for each spinning machine to guide the operator to tend to the machine with the greatest number of active breakages. The first method offered is that of installing 4 signal lights (2 per side) for each spinning machine that can indicate the range of active breakages. In this case, the operator will take action first on the frames with the red signal, then on the yellow, and finally on the green.



The second method requires installation of 1 or 2 LCD display 16x24 that shows the number of breakage per side. The LED colour can be green (no action needed) or red (operation action needed)

Roving Stop



Roving Stop is an electromagnetic device that stops the feeding of the roving when the sensors detect a breakage of yarn or filament. This is one of the key product of Pinter group. It's an optional tool, and can be linked to any of our sensors: electromagnetic or Optifil, for spindle monitoring or for Core yarn device. Each device is connected to a special rail, where an electronic PCB continuously monitors the communication with the sensors, and

when a new yarn (or filament) breaks, it activates the corresponding roving stop, preventing lapping and roving waste. From control unit Tbox, it's easy to modify the settings of the Roving stops.

Control Box

One for each ring frame, with Pinter's unique Powerspin4 to supply power to the spindle monitoring system, and collect all the data of breakages and low twists etc, after collecting all the data, it will submit to server by intranet, customer can achieve the target of statistic and high efficiency management.

- a. Per machine data of 20 shifts will be recorded and saved. No data loss due to power failure.
- b. In case of server crashed, the single ring frame will run normally.
- c. In case of single ring frame has emergency shut down, other ring frames and server will run normally.

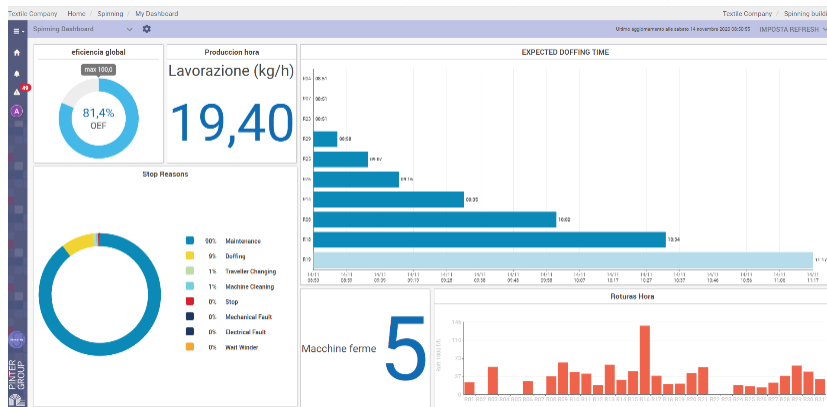


Smart Dragon Monitoring

Production monitoring

SD-Monitoring is the innovation on the textile monitoring software sector. Completely developed as a Web platform, it can work on mill internal server or in the cloud.

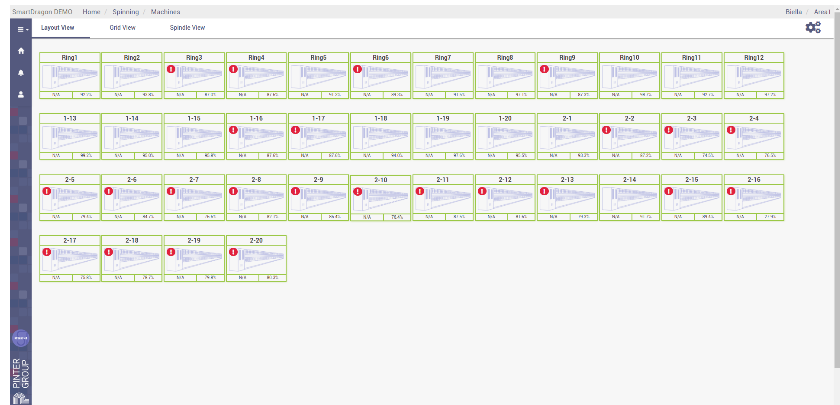
The SD-Monitoring system is designed to monitor and plan production in Carding, Combing, Blending, Recombing, Preparation, Spinning, Winding and Twisting.



for the machines stops.

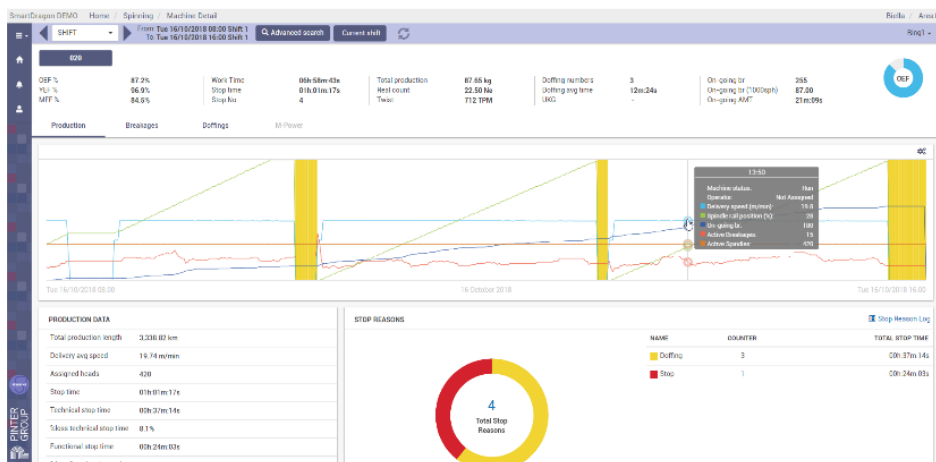
Another screen shows the real time status of each machine and can be configured for graphic or data mode.

In graphic mode, an overview of the system is shown with representations of the machines and departments to be monitored.



In data mode, a grid is shown for each department, and the main production data and the Department Alarms are reported.

In both ways of viewing your system, the most important data – for example the active stop reason, the current batch being processed, efficiency and the speed of the machine – are highlighted.

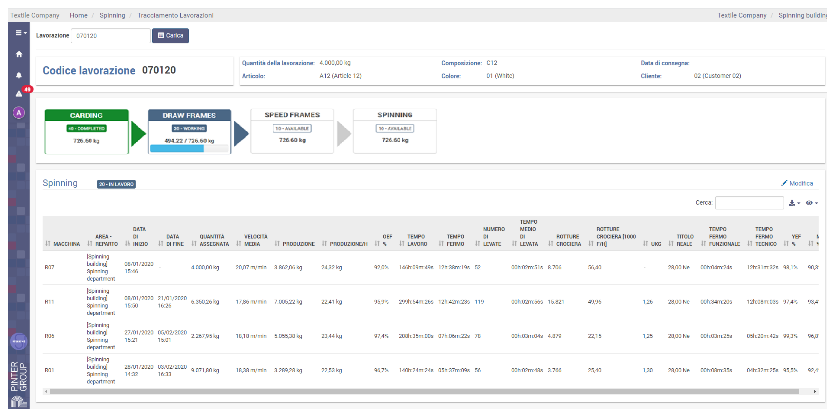


Select a machine to see

detailed data with all the monitored data totalled by shift. A minute-by-minute graphic allows analysis of the machine's progress, evaluating speed, stoppages and stop reasons for each minute of operation. Part of the screen shows a downtime summary, indicating the reason and the number and duration of stoppages. The production

summary shows kg and km produced, downtime and worktime with related yields. Downtime and yields are divided between generic (counting all downtime for a shift), technical (counting only stoppages due to processing) and operative (counting only stoppages due to the department's organization).

A Production Progress management screen tells the user the state of progress in the processing within the production cycle.



Analysis can then be done by evaluating various yields, production, the number and average time of doffing, breakages in 1000 f/h, etc.

Another screen provides an overview of the department, graphically showing the stoppages for various machines on the shift.

To enter stop reasons, an operator code is requested in order to monitor operator efficiency in the department.



Breakage control methods

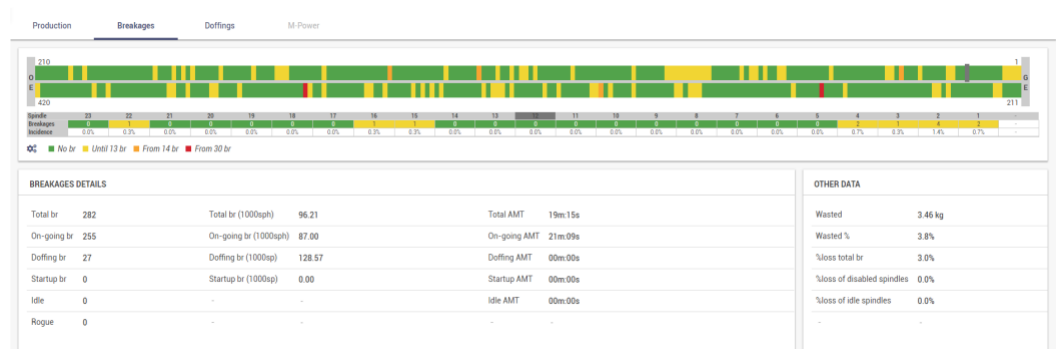
The *SD-Monitoring* system has the goal of making available the reported data on breakages, spindle by spindle. To this end, a sensor is installed for each spindle that is able to report possible breakages of threads or the twists. The terminal continuously monitors the sensors, making available in real time the current breakages in the machine. A signal light (or an LED cube) is installed in each side of the spinning machine, to guide the operator in taking quick and targeted action. To further facilitate the operator's repairs, every single sensor has a bright red LED that signals current breakages or low-twist spindles.

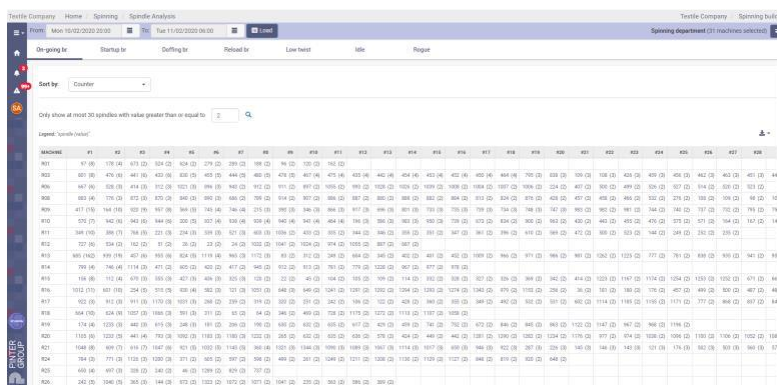
The software provides real-time information on the progress of the spinning machines, and by setting ranges for the alarms tied to the article in production, operator warnings are activated reporting 1000 f/h breakages to allow immediate action by the technician.

Displays of overviews of breakages are made available, mapping them to the side of the machine, allowing identification of possible anomalous

areas, perhaps due to mechanical problems in the spinning machines or due to broken bobbins.

The reported breakages are separated by production breakages, breakages in doffing, reloading breakages and start-up





displays and reports, the average time for stopped spindles is reported, tied to the operator assigned to the machine.

A special display shows production breakages occurring for each doffing, and breaks down these breakages by side and ring rail position; this permits in-depth analysis of the distribution of breakages during doffing and allows making necessary changes to production speed.

In the on-board terminal on the machine, an alarm indicates when a spindle exceeds a pre-set limit for breakages, allowing action to be taken on possible problems with slips, broken bobbins or other issues.



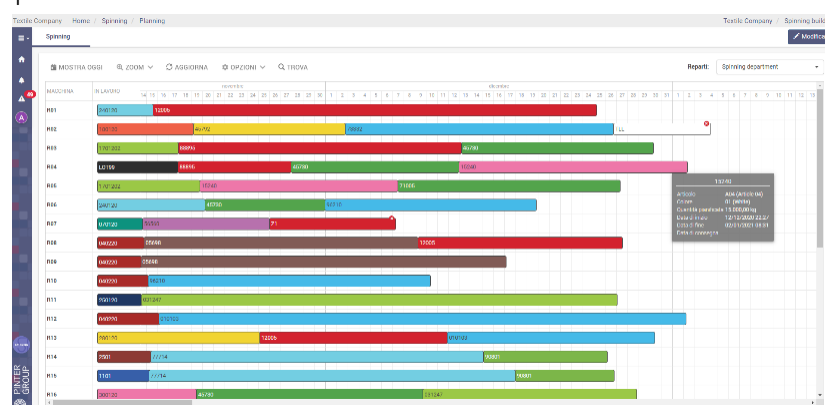
Production planning

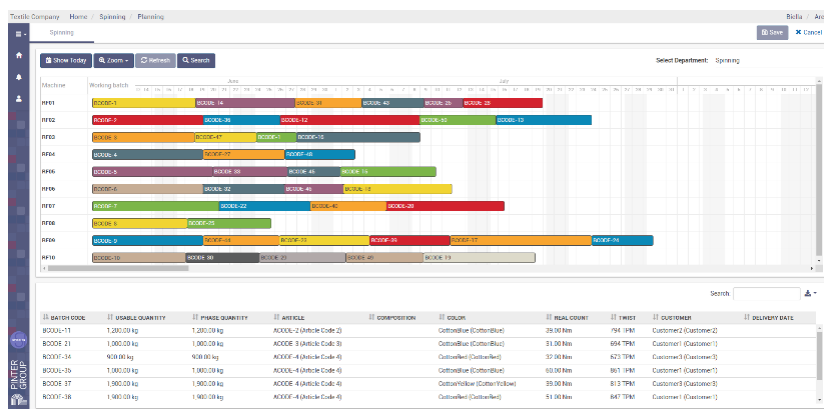
The term *production planning* means the activity needed to organize the productive process in order to optimize the use of production personnel and equipment.

In the SD-Monitoring system, production planning is managed by using a Gantt chart. The chart is a visual representation of each process in a coloured bar proportional to the duration of the process on that machine. Multiple bars are associated horizontally with a given machine in order of priority that the user defines.

The length of the process is calculated based on historical speed and efficiency for that article. The SD-Monitoring system's planning projects this information onto the work calendar for the department, determining the length of the segment and its relative chronological placement (date and time of beginning and end of the process).

The selection of processes to plan is done by drawing from data imported from the company's information system, by choosing from a list or searching for a process code. Importing can be done automatically (e.g., every day at a set time) or manually (upon request by the operator); the addition of new processes may also be done manually.





Switching between planning for one department to another department is easy, allowing the manager to keep an eye on the interactions of the scheduling between the various departments and to control for possible warnings that are issued automatically when the software identifies something “inconsistent” between the unloading data for one department and the loading data for other departments.

The main screen for the planning module shows the machines with a predicted length of duration that can be set by the user. A detail window displays the most meaningful data for the process in question (Code, Article, Loading and Unloading Data, Title or Fabric Weight, Client, etc.). Access to scheduling is protected by password. By selecting processes using only the mouse (the drag-and-drop method), those processes may be assigned to machines. Machine stoppages are also automatically scheduled to correspond with related batch changes or can be entered manually for any reason.

Alarms are automatically generated when delivery dates are overrun or when the estimated time for material preparation does not match the work start date at the spinning machine.

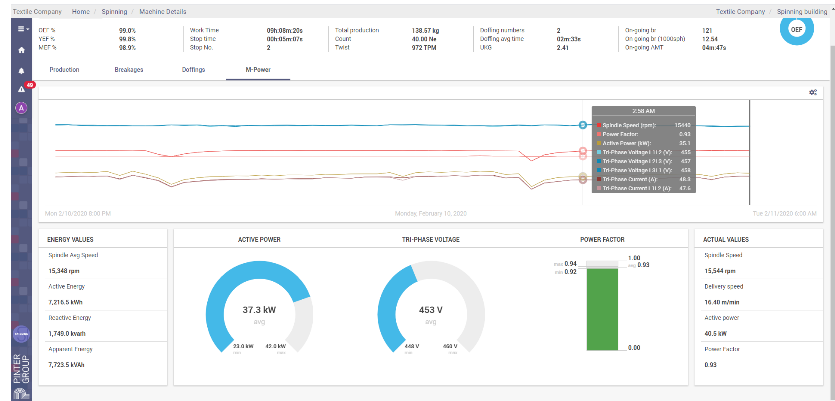
Additional modules (optional)

M-Power - monitoring of electrical costs of machines

M-power collects the electrical cost data of the machines through devices installed in the three-phase power supply.

The data is processed and transformed into the following information:

- Kw / Kgs costs produced by each machine
- Instantaneous voltage of each machine
- Instantaneous amperage of each machine
- Total energy costs of each machine
- Power factor of each machine
- Active energy measured in kWh.
- Reactive energy measured in kVarh.

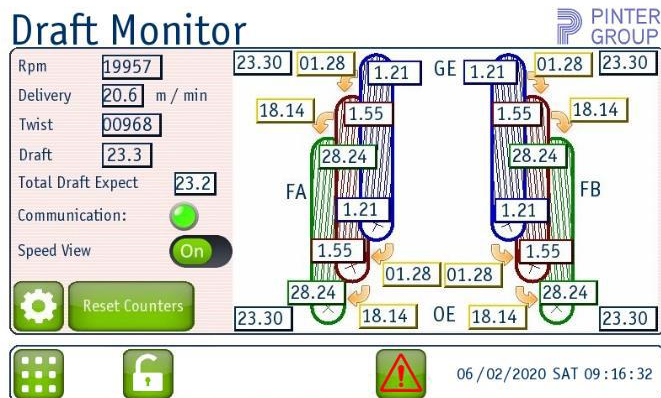


Reactive energy causes an overload in lines generating additional and significant costs in the bill.

- Power factor ($\cos \phi$): relates the active and apparent energy consumption of an installation. The apparent energy in turn it depends on the active and reactive energy. For the same active energy consumption, the higher the energy consumption

M-DRAFT – monitoring of the draft with alarms

Draft Monitor checks the draft at the beginning and at the end of the spinning machine and reads: 3 drafting each side (1°-2° cylinder, 2°-3° cylinder, 1°-3° cylinder) and 3 speed each side (1° cylinder, 2° cylinder, 3° cylinder).



RPM: RPM read from Yarn Sensor or from proximity sensor WB as set in Machine Inputs.

Delivery (Speed): Show the actual delivery speed.

Twist: Show the actual Twist calculated on delivery speed and RPM.

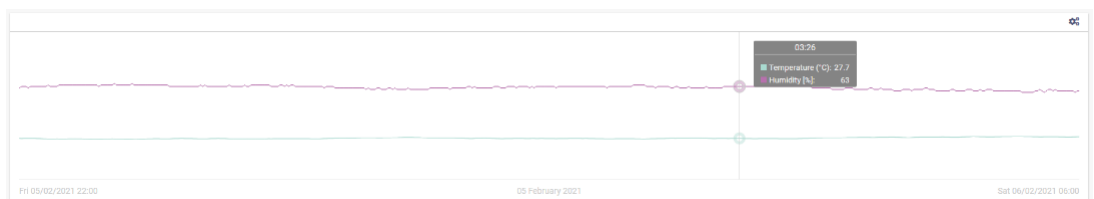
Draft: Roving draft calculated on all sensors connected with Draft Monitor.

Total Draft Expected: value used to compare the values read from the sensor. If the values read are different from the value expected appear an alarm.

Total Draft Expected set as 0: the alarm activated from the difference between the values read and the average of the same values

M-CLIMA – Temperature and Humidity monitoring

This module monitoring in real time the temperature and humidity in the department. This data are directly connected whit the monitoring system software. The values are graphically represented minute by minute and are also reported to compare the trend of breakages with the instantaneous temperature and humidity.



M-TV – Big department display



Monitor the trend through a TV screen hung on a wall on your department. In this big display is possible to view trend graphs, list of stopped machines, number of active breaks for each spinning machine, intervention times for operator and other widgets that can be configured directly by the customer. it is possible to have up to 10 timed screens in rotation.

Future Upgradation for Automation

Start from 2019, Pinter group started the project which we've been focusing long time ago, which is APR (Automatic Piecing Robot), with the unstopped hard work from three R&D teams located in Spain, Italy, and China, we've showcased our APR during ITMA Shanghai 2020 in year 2021, and the most updated version will be presented during ITMA Milano, 2023, and which will be commercialized soon afterwards.

This APR is based on the same platform of our spindle monitoring system which we've been keeping improving for the last 40 years, once the APR is ready, you can upgrade your spindle monitoring system seamlessly to spindle monitoring system with APR.



Future Upgradation for a Close Circle with Winding Machine

In 2023, Pinter Group and Uster Technologies signed a cooperation agreement that puts profitable spinning as top priority. Both companies are convinced that greater value can be created from combining efforts and solutions. The agreement includes close cooperation and customer support in the field of Ring Spinning Optimization.

Furthermore, the agreement foresees that the Uster RSO 3D feature will be available for customers using Pinter's Optifil ring monitoring system in combination with Uster Quantum yarn clearers. This step heralds the era of preventive yarn clearing and expands the RSO 3D benefits to a larger market.

The extended collaboration includes compatibility between Uster Sentinel, Optifil or Pinter's core yarn attachments using common roving stop equipment and connectivity to the Uster platform.

Uster RSO 3D increases profitability with direct process optimization and traceability from ring spinning to winding.



Proforma Invoice / Offer

Description	Q.ty	Unit Price (USD)	Price (USD)
18 sets Rieter G32 X 1.824 ring frame, totally 32.832 spindles			
Optifil +Roving stop for n.18 G32 RING FRAME with 1.824 spindles, gauge 70mm Individual yarn optical breakage detection system Optifil Equipped with a red LED to signalized the breakage. Adaption of the system on the ring frame on the ringrail. When connected to the monitoring system is able to read the twist of yarn as well.(High Light version) Roving stop device that stops the top back bottom roller and therefore breakes the roving between the middle and back bottom roller. The Roving stop may be triggered by any of the availble filament or yarn sensors, in this case, by Optifil sensors	18	10,032.00	180,576.00
ST-BOX central electronic control unit for n.18 G32 RING FRAME with 1.824 spindles, gauge 70mm It supplies power to the spindle monitoring system through the PowerSpin 4. It also includes the board CPU T4 that collects breakage information and spindles with low RPM. It acquires sensor data and delivers the information to the Spining Monitoring software (if purchased) for its analysis. Data transfer via Ethernet cable.	18	Included	Included
7" Colour Touch Screen for n.18 G32 RING FRAME with 1.824 spindles, gauge 70mm Shows the data and allow to change the working parameters of the system. Remote control funcation available.	18	Included	Included
LED Display (double face) for n.18 G32 RING FRAME with 1.824 spindles, gauge 70mm	36	Included	Included
Lamp tower for n.18 G32 RING FRAME with 1.824 spindles, gauge 70mm	72	Included	Included
M-Power for Ring section	18	250.00	4,500.00
M-Clima for ring section Installed on OE, next GE position in sequence, every sensor shared by 2 neighbor machines	18	250.00	4,500.00
M-Draft for ring frame	18	250.00	4,500.00

M-TV for ring area This is the M-TV module to connect to the TV, customer need to buy the TV to connect to the M-TV module	1	500,00	500
SMART DRAGON Web based Monitoring software for the Spinning department , comprising: 1) Dashboard 2) Layout of Machines 3) Machine Details 4) Stop Chart 5) The Breakage/Speed chart 6) The Lot Tracking 7) The Spindle Report 8) App Spinning for your mobile (Android and iOS) 9) This version includes the production basic module with the screens described above.	1	25.000,00	Included
Total EX-works Pinter Factory			194,576.00
Freight Charge			1,750,00
Insurance			214.03
Total CIF Haiphong			196,540.03

Terms and Condition

Validity of the offer:	1 month
Date of shipment of the goods:	30 days after 30% T/T down payment received.
Payment terms:	30% T/T down payment 70% balance payment before shipment
Bank Information:	
Account with Bank:	BENE BANK: SHANGHAI PUDONG DEVELOPMENT BANK,QINGDAO BRANCH ADD.: NO.53 WEST HONG KONG ROAD,QINGDAO,SHANDONG 266071,CHINA SWIFT: SPDBCNSH170 BENE'S A/C NO.: 69011454700000816
Beneficiary:	NAME: QINGDAO FORTEX INTERNATIONAL CO., LTD. ADD.: ROOM 205, UNIT 2, BLOCK 1, NO. 26, JINGDE ROAD, QINGDAO, CHINA. TEL: +86-532-68677786
Installation cost:	Pinter will pay the round ticket(if any) for the technician, customer will pay boarding and lodging, local transportation for the technician.
Material not included:	- Cable for tower lamps or other signaling method - Customized support for tower lamps or other signaling method - Ethernet network to connect each STBOX to computer server - Cable for LED display - TV screen - TV supports - Wiring from server to TV
Warranty:	- Warranty period is on years from the running test. If installation and running test will be delayed due to customer installation schedule, customer will store goods in safe and appropriate way. - During warranty period, out of order parts will be replaced free of charge. - Pinter Group's warranty covers only those defects which arise as a result of normal use of the product, and do not apply to any improper or inadequate use, maintenance or modification.